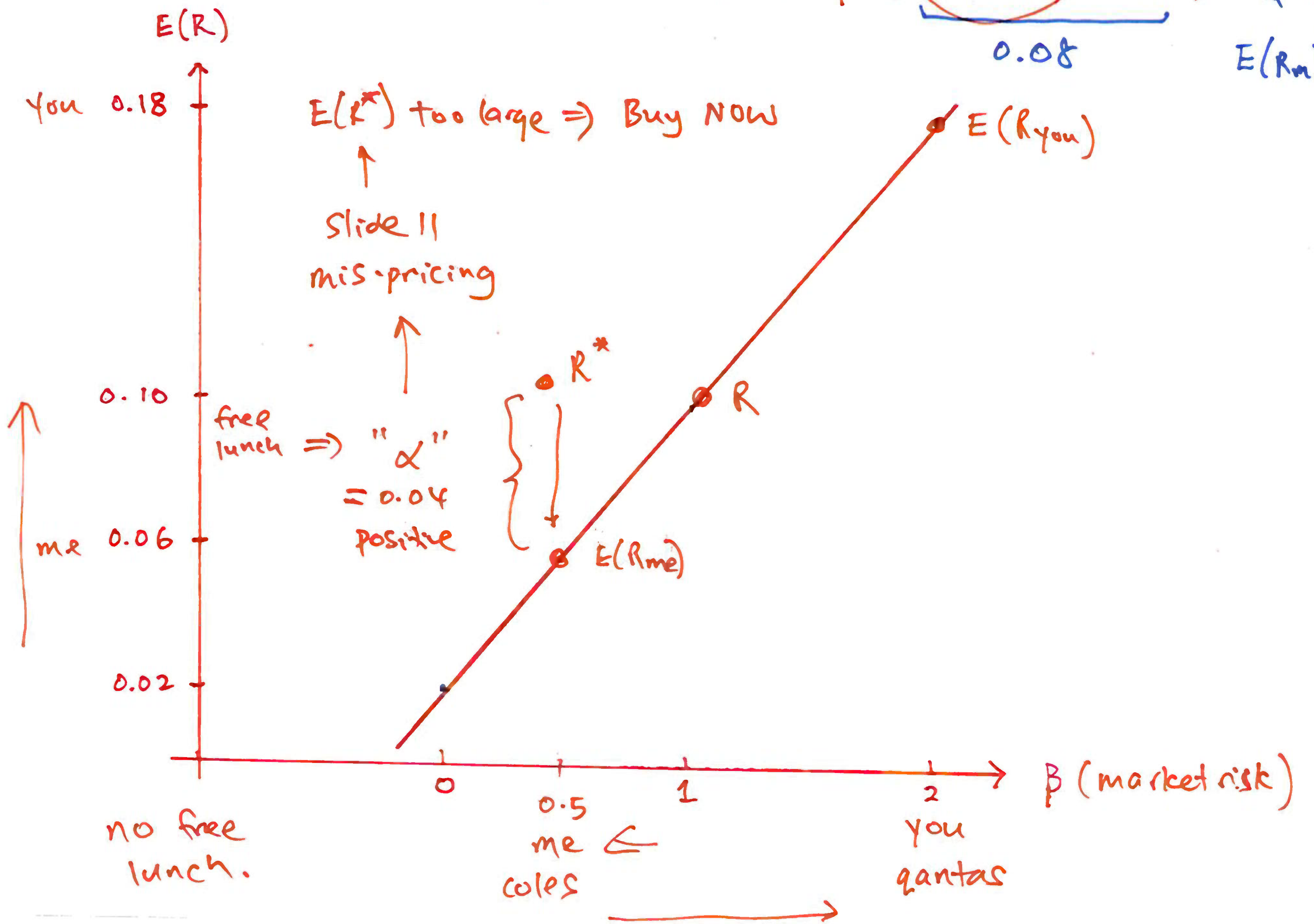
Week 5: Asset Pricing Models

Capital Asset Pricing Model

$$E(R) = r_f + \beta\,(\underbrace{E(R_m) - r_f}_{0.08})$$

$r_f = 0.02$

$E(R_m) = 0.10$



Idiosyncratic risk → specific to a company not other companies

e.g. 9% per month for A
vs
2% " " for B
↓
But return for company A should not be higher than B
↓
Why?

Idiosyncratic risk derives from the fact that many of the dangers that surround an individual company are peculiar to that company and, perhaps, its immediate competitors.

Idiosyncratic risk can therefore be eliminated by holding a well-diversified portfolio.

But there is also some risk that cannot be avoided no matter how much one diversifies. This risk is generally known as market risk.

Market risk affects everyone

Market risk is sometimes called the **systematic risk** .

Market risk derives from the fact that there are other economywide and global dangers and opportunities that are faced by all businesses.

unemployment ↑

The fact that stocks have a tendency to "move together" reflects the presence of market risk, risk that cannot be eliminated through diversification.

In CAPM, market risk is captured by **beta** (β).

CAPM

The CAPM pricing formula:

$$\mathbb{E}(R_A) - r_f = \beta^A \times \left(\mathbb{E}(R_M) - r_f\right)$$

When the market risk premium goes up by 1 percentage point (e.g. from 2% to 3%), the asset's expected excess return increases by β^A percentage points.

If $\beta^A = 2$, a 1% decrease in the market risk premium is expected to lead to a 2% decrease in the asset A's expected excess return, on average. Such assets are relatively risky.

Market risk

In CAPM, market (systematic) risk is captured by **beta** (β).

A stock's beta measures its sensitivity to movements in the overall market:

qantas Aggressive (higher risk, higher return) : $\beta > 1$ no free lunch

Tracks the market : $\beta = 1$

coles. Conservative (lower risk, lower return) : $0 < \beta < 1$

Independent of the market : $\beta = 0$

Priced ⇒ no free lunch.

Priced means that a type of risk affects an asset's expected return.

If a risk is priced, investors require higher expected return as compensation for bearing more of that risk.

Market risk is priced since investors must bear this economy-wide risk, they require compensation for it.

As a result, assets with greater exposure to market risk generally have higher expected returns.

idio is not priced.

CAPM: Assumptions

The CAPM assumes that each investor holds a so-called mean variance efficient portfolio, a portfolio that gives maximum expected return for a given variance (level of risk).

If all investors hold the same beliefs about expected returns and variances of individual assets, and in the absence of transaction costs, taxes and trading restrictions of any kind, it is also the case that the aggregate of all individual portfolios, the market portfolio, is mean variance efficient.

CAPM Regression Model Setup

$t = 1, \ldots, T$

1990 2000

Running regression to estimate the CAPM beta:

$$r_{it} - r_{ft} = \alpha_i + \beta_i \left(r_{mt} - r_{ft} \right) + \epsilon_{it}$$

There are two numbers to be estimated: α_i and β_i.

CAPM Regression Model Setup

- Y Let r_{it} be the return on an individual asset at time t.
- Let r_{ft} be the return on a riskless asset at time t.
- X Let r_{mt} be the return on "the market" at time t.
- In many cases, we will simply let r_{ft} be the return on a short-term treasury bond, usually 1 or 3-month.
- However, in practice r_{mt} is taken to be a weighted portfolio of all market activity.

Interpretation of Alpha

The CAPM says that $\alpha_i = 0$ but by allowing $\alpha_i \neq 0$, we recognize the possibility of mispricing.

If α_i is nonzero, then the asset is mispriced, at least according to the CAPM.

If $\alpha_i > 0$ then the returns are too large on average. This is an indication of an asset worth purchasing.

CAPM Interpretation: α_i

α-risk refers to an assets ability to earn abnormal returns relative to the market return.

We can classify assets according to their α-risk:

$$
\begin{aligned}
\text{Inadequate Reward for assumed risk} &: \alpha_i < 0 \quad \Rightarrow \text{sell now}\\
\text{Adequate Reward for assumed risk} &: \alpha_i = 0\\
\text{Excess Reward for assumed risk} &: \alpha_i > 0 \quad \Rightarrow \text{Buy now}
\end{aligned}
$$

The CAPM as a **regression model**:

affect everyone

$$r_{it} - r_{ft} = \alpha_i + \beta_i \left(r_{mt} - r_{ft} \right) + \epsilon_{it}$$

We can think of r_{mt} as capturing "market-wide" news at time t that is common to all assets, and β_i captures the sensitivity or exposure of asset i to this market-wide news.

An example of market-wide news is the release of information by the government about the national unemployment rate.

If the news is good then we might expect r_{mt} to increase because general business conditions are good. Different assets will respond differently to this news and this differential impact is captured by β_i.

Total risk $\Rightarrow$ market risk + idio risk

The CAPM as a **regression model**:

for Asset "i"

not Asset "j"

$$r_{it} - r_{ft} = \alpha_i + \beta_i \left(r_{mt} - r_{ft} \right) + \epsilon_{it}$$

capture "idio" risk

Because ϵ_{it} is assumed to be independent of r_{mt}, we can think of ϵ_{it} as capturing specific news to asset i that is unrelated to market news or to specific news to any other asset j.

Examples of asset-specific news are company earnings reports and reports about corporate restructuring (e.g., CEO resigning).

Hence, specific news for asset i only effects the return on asset i and not the return on any other asset j.

Total risk of r_{it}

For simplicity, we assume the following CAPM as a regression model

$$r_{it} - r_f = \beta_i \left(r_{mt} - r_f\right) + \epsilon_{it}$$

where the risk-free rate r_f is time-invariant, the asset is correctly priced such that $\alpha_i = 0$, $E(\epsilon_{it}) = 0$ and $\text{cov}(r_{mt} - r_f, \epsilon_{it}) = 0$.

The total risk of $r_{it} - r_f$ is thus

$$\underbrace{\text{var}(r_{it} - r_f)}_{\text{total risk}} = \underbrace{\beta_i^2 \text{var}(r_{mt} - r_f)}_{\text{market risk}} + \underbrace{\text{var}(\epsilon_{it})}_{\text{idio risk}}. + \underbrace{2\,\text{cov}(r_{mt} - r_f, \epsilon_{it})}_{= 0}$$

total risk ↑ var $(r_{it} - r_f)$

no relationship

Total risk of r_{it}

$$\mathrm{var}(r_{it} - r_f) = \beta_i^2 \,\mathrm{var}(r_{mt} - r_f) + \mathrm{var}(\epsilon_{it}).$$

This shows that the variance of the return on a stock consists of two parts: a part related to the variance of the market index and an idiosyncratic part.

In economic terms, this says that total risk equals market risk plus idiosyncratic risk.

$$\mathrm{var}(r_{it} - r_f) = \beta_i^2 \,\mathrm{var}(r_{mt} - r_f) + \mathrm{var}(\epsilon_{it}).$$

The R^2, being the proportion of explained variation in total variation,

$$R^2 = \frac{\beta_i^2 \,\mathrm{var}(r_{mt} - r_f)}{\mathrm{var}(r_{it} - r_f)} = \frac{\text{market risk}}{\text{total risk}}$$

✓ R^2 provides an estimate of the proportion of the total risk that is due to systematic risk.

$1 - R^2$ gives us the ratio of how much of the variance in r_{it} comes from the idiosyncratic component.

For example, if $R^2 = 0.598$, it is estimated that 59.8% of the risk (variance) of the stock is due to the market as a whole, while 40.2% is idiosyncratic (firm-specific) risk.

Idiosyncratic Risk

The standard error of the regression (SER) is an estimator of the standard deviation of the regression error ϵ_{it}.

$\hat{\varepsilon}_1 \rightarrow$ first obs

$\hat{\varepsilon}_T$ - last obs

$\downarrow$

for Asset i

not Asset j

The SER is computed using OLS residuals $\hat{\epsilon}_{it}$; that is,

$$SER = \sqrt{\frac{1}{T-K-1}\sum_{t=1}^{T}\hat{\epsilon}_{it}^2}$$

$\Rightarrow$ same unit as data (Y)

where K is the number of regressors in the model.

In R, SER is referred to as the '**Residual standard error**'.

Thus, SER estimates the idiosyncratic risk. $\Rightarrow$ 9% for A

CAPM: Market Factor

- To implement the CAPM, we require some notion of "the market factor", which is the excess returns on the market portfolio.

- For our purposes, we will take as the market factor the CRSP (center for research in securities prices) portfolio.

- The CRSP index includes stocks from all the firms listed in the NYSE, AMEX, and NASDAQ indices. As such, it is a good proxy for "the market factor".

- Sometimes practitioners use S&P 500 index as the market factor.

CAPM Estimation

- Define $y_{it} = r_{it} - r_{ft}$ and $x_t = r_{mt} - r_{ft}$.
- Given data on $\{y_{it}, x_t\}$ for an asset i through time, $t = 1, 2, ..., T$.
- Population parameters α_i and β_i estimated using OLS.
- Minimize

$$\sum_{t=1}^{T} (y_{it} - \alpha_i - \beta_i x_t)^2.$$

- Well-known OLS solutions

$$\hat{\alpha}_i = \bar{y}_i - \hat{\beta}_i \bar{x},$$

$$\hat{\beta}_i = \frac{\sum_{t=1}^{T} (x_t - \bar{x})(y_{it} - \bar{y}_i)}{\sum_{t=1}^{T} (x_t - \bar{x})^2}.$$

CAPM regression

Consider the estimation of the CAPM regression for Microsoft (msft) using monthly data from January 1990 to Dec 2000.

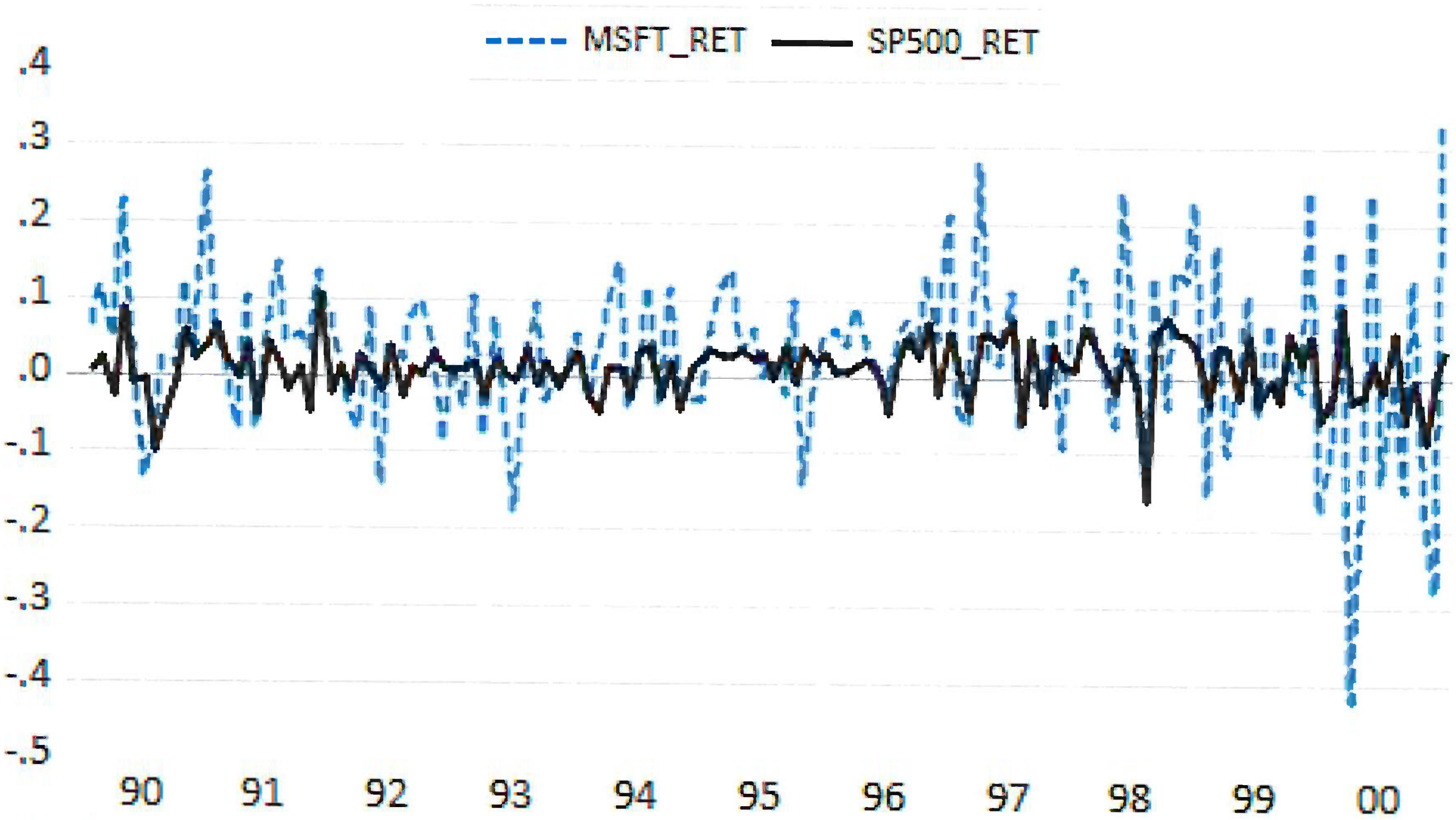
The S&P 500 index is used for the market proxy, and the 30-day T-bill rate is used for the risk-free rate.

```
> tail(data.xts)
            $  MSFT     RF    SP500 index
Aug 2000 69.8125 0.0048 1517.68
Sep 2000 60.3125 0.0048 1436.51
Oct 2000 68.8750 0.0048 1429.40
Nov 2000 57.3750 0.0048 1314.95
Dec 2000 43.3750 0.0048 1320.28
Jan 2001 61.0625     NA 1366.01

> # merge two series msft and sp500
> msft.sp500 <- merge(data.xts$MSFT,data.xts$SP500)

> # log return
> lr = na.omit(diff(log(msft.sp500)))

> head(lr)
                MSFT        SP500
Feb 1990  0.06537980  0.008502706
Mar 1990  0.11470787  0.023965543
Apr 1990  0.04630427 -0.027255168
May 1990  0.23003429  0.088000911
Jun 1990  0.04025768 -0.008926024
Jul 1990 -0.13353816 -0.005236860
```

not %

decimal.

Monthly log returns on the S&P 500 Index and msft.



not %

```
Call:
lm(formula = (lr$MSFT - rf) ~ I(lr$SP500 - rf))
```
excess excess

```
Residuals:
     Min        1Q    Median        3Q       Max
-0.38355 -0.05661  0.00234  0.06036  0.19906
```

Coefficients: $\hat{\alpha}$ ⇒ not in % ⇒ $\hat{\alpha} > 0$ ⇒ 1.2% per month.

```
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.012817   0.007995   1.603    0.111
I(lr$SP500 - rf) 1.525871   0.199845   7.635 4.44e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$\hat{\beta}$

→ not %

```
Residual standard error: 0.09027 on 129 degrees of freedom
Multiple R-squared:  0.3113,    Adjusted R-squared:  0.3059
F-statistic:  58.3 on 1 and 129 DF,  p-value: 4.435e-12
```

SER

3 R^2

CAPM regression

1

The estimated value for β for Microsoft is 1.526 with an estimated standard error of 0.2. So, Microsoft is judged to be riskier than the S&P 500 index.

2

The estimated value of α is 0.013 with an estimated standard error of 0.008. This implies that Microsoft stock is expected to have a return that is 1.3% per month higher than the CAPM predicts.

3

The percentage of market risk is $R^2 = 0.31$ and the percentage of idiosyncratic risk is $1 - R^2 = 0.69$.

The estimated magnitude of idiosyncratic risk is = 0.090 or 9% per month.

Additional Risk Factors: Multi-Factor CAPM

salary educ

Q1 : $Y_1 = \beta_0 + \beta_1 X_1 + \varepsilon_i$

different

house price size

Q2 : $Y_2 = \beta_0 + \beta_1 X_2 + \varepsilon_i$

ME

Q1 : qantas

factor

$r_{it} = \beta_0 + \beta_1 X + \varepsilon$

same

Q2 : coles

$r_{jt} = \beta_0 + \beta_1 X + \varepsilon$

loading ⇒ slide 36

exposure to X

Alternatives to CAPM

- The CAPM model has been extensively studied.
- Alternatives to the standard CAPM model are:
  1. the Fama-French 3 factor model
  2. the Fama-French 4 factor model

portfolio ⇒ manuf industry.

Fama-French 3 Factor Model

- Fama and French (1993) have proposed to augment the CAPM model by including two additional risk factors to explain investment returns.

- Colloquially these factors are referred to as ① **size factor** and **value factor**. ②

Fama-French 3 Factor Model

- **Size** refers to the market capitalization.
- **Size** factor, or “Small minus big” (**SMB**) accounts for the spread in returns between small- and large-sized firms.
- Thus **SMB** refers to the performance of big stocks versus that of small stocks.
- Comes from observation that firms with small market cap outperform large market cap firms.

SMB > 0

Fama-French 3 Factor Model

- **Value factor**, or **High-Minus-Low**, (**HML**), is an additional factor that captures the performance of "value" stocks relative to growth stocks.

- **HML** factor accounts for the spread in returns between value and growth stocks.

- HLM factor argues that companies with high book-to-market ratios, also known as value stocks, outperform those with lower book-to-market values, known as growth stocks.

$HML > 0$

Fama-French 3 Factor Model

Taken together with the "market factor", SMB and HML form the multi-factor, or three-factor, CAPM:

loading.

$$r_{it} - r_{ft} = \alpha_i + \beta_{i1}(r_{mt} - r_{ft}) + \beta_{i2}\underline{\mathrm{SMB}}_t + \beta_{i3}\underline{\mathrm{HML}}_t + \epsilon_{it}.$$

factor 1 2 3

Sometimes called the Fama-French 3 factor (FF3F) model.

The coefficients on SMB and HML tell you how sensitive the asset or portfolio is to those factor returns.

SMB coefficient

size factor

Interpretation of the SMB coefficient

$$\beta_{i2}$$

measures the asset's size exposure.

Co-moving with SMB

- If $\beta_{i2} > 0$, the asset behaves more like a small-stock portfolio.
- If $\beta_{i2} < 0$, the asset behaves more like a big-stock portfolio.
- If $\beta_{i2} \approx 0$, the asset has little systematic tilt towards either small or big firms.

SMB > 0

Because SMB = Small minus Big, a positive coefficient means when small stocks outperform big stocks, this asset also tends to do relatively well, holding the other factors fixed.

SMB coefficient

Example: If

$$\beta_{i2} = 0.40$$

then a 1% increase in SMB is associated with about a 0.40% increase in the asset's excess return, all else equal.

So we would say the asset has a positive small-cap tilt.

HML coefficient

1:05
1:10 back

Interpretation of the HML coefficient

$$\beta_{i3}$$

measures the asset's value exposure.

co-move positively with HML > 0

- If $\beta_{i3} > 0$, the asset behaves more like a value stock or portfolio.
- If $\beta_{i3} < 0$, the asset behaves more like a growth stock or portfolio.

HML > 0

- If $\beta_{i3} \approx 0$, the asset has little systematic tilt towards value or growth.

Because HML = High minus Low, a positive coefficient means when value stocks outperform growth stocks, this asset also tends to do relatively well, holding the other factors fixed.

HML coefficient

Example: If

$$\beta_{i3} = -0.60$$

then a 1% increase in HML is associated with about a 0.60% decrease in the asset's excess return, all else equal.

So we would say the asset has a growth tilt.

Very simple intuition

- Positive SMB loading $\rightarrow$ small-cap flavour
- Negative SMB loading $\rightarrow$ large-cap flavour
- Positive HML loading $\rightarrow$ value flavour
- Negative HML loading $\rightarrow$ growth flavour

Important caution:

These coefficients do not mean the stock is literally small or literally a value stock. They mean its returns co-move with those factor portfolios.

So the interpretation is about factor exposure, not just the firm's label.

Fama-French 3 factor

Example

- Using the Fama-French data it is easy to estimate the multi-factor CAPM regression model.
- Commonly done for portfolios, so we will consider some new portfolios.
- Industry portfolios are formed from stocks of different firms collected according to their four-digit SIC.
- The portfolios are roughly in five categories: consumers, manufacturing, technology, health and Other.
- Let's use monthly returns on the manufacturing and technology portfolios from 1926m8 to 2017m10.

Ken ~~Fresh~~ French website:

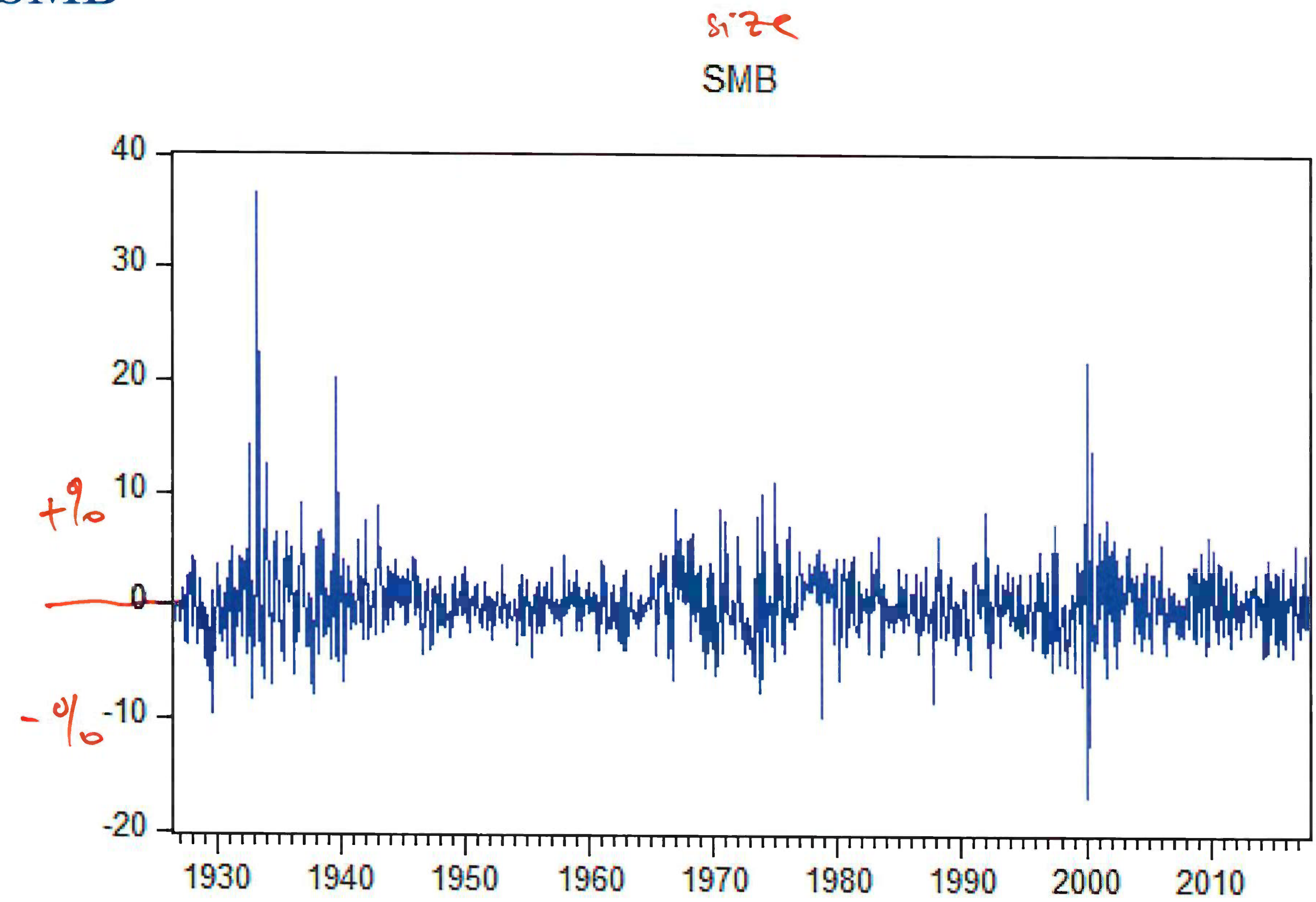
return

tech

% % % %

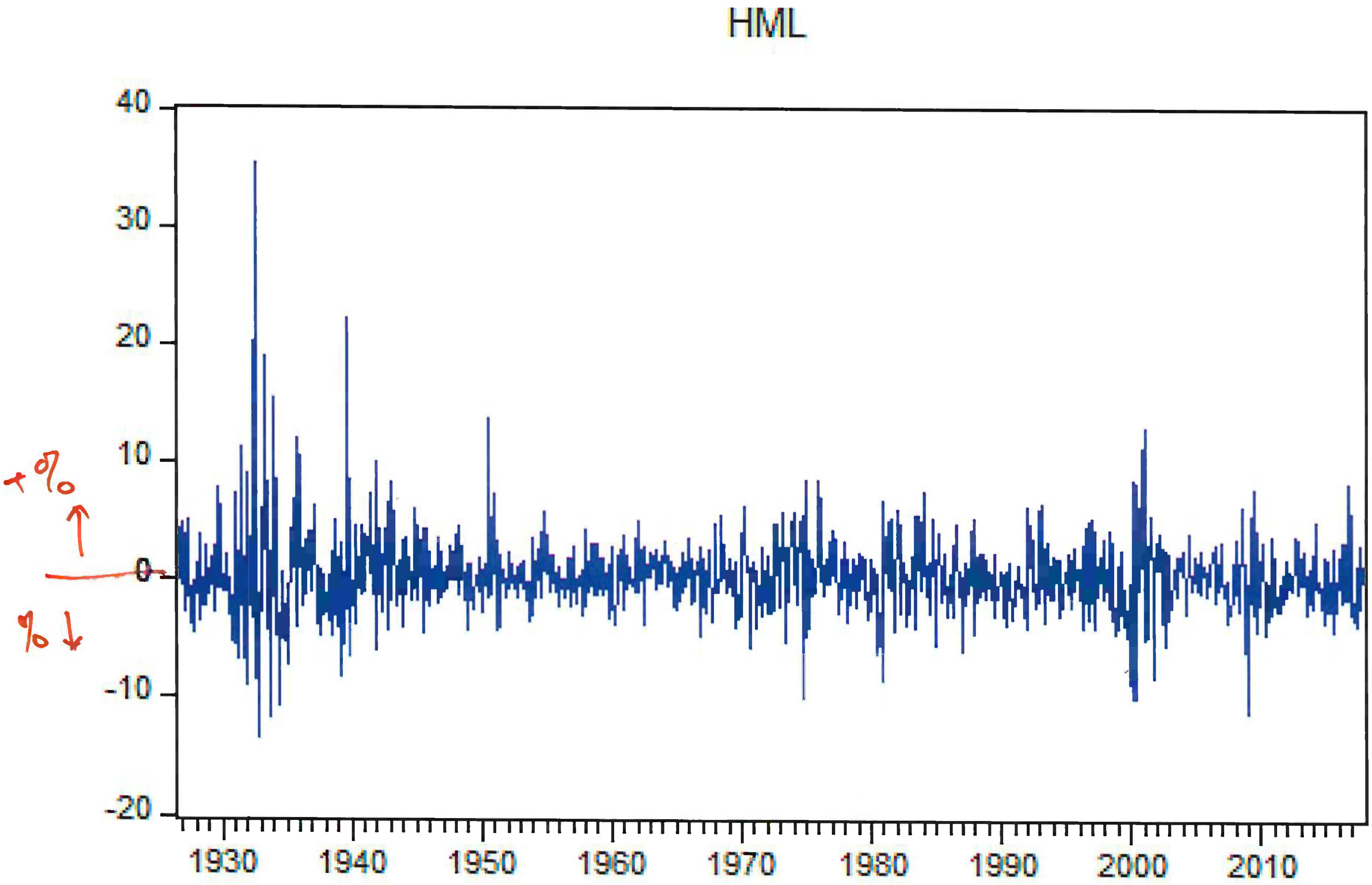
```
> head(data.xts)
         HITEC MANUF   RF   HML   SMB MKT_RF
Aug 1926  2.27  2.27 0.25  4.19 -1.40   2.64
Sep 1926  0.72 -0.33 0.23  0.01 -1.32   0.36
Oct 1926 -4.70 -2.92 0.32  0.51  0.04  -3.24
Nov 1926  1.36  2.09 0.31 -0.35 -0.20   2.53
Dec 1926  1.49  3.60 0.28 -0.02 -0.04   2.62
Jan 1927  0.38  1.37 0.25  4.83 -0.56  -0.06
> tail(data.xts)
         HITEC MANUF   RF   HML   SMB MKT_RF
May 2017  1.49 -2.57 0.06 -3.75 -2.54   1.06
Jun 2017  1.21  0.76 0.06  1.32  2.15   0.78
Jul 2017  1.49  0.86 0.07 -0.28 -1.42   1.87
Aug 2017 -0.81 -3.70 0.09 -2.26 -1.71   0.16
Sep 2017  5.67  8.79 0.09  3.02  4.53   2.51
Oct 2017  1.52  0.75 0.09 -0.09 -1.94   2.25
```

SMB



size
SMB
40
30
20
10
0
-10
-20
+%
-%
1930
1940
1950
1960
1970
1980
1990
2000
2010

HML


HML
40
30
20
10
0
-10
-20
1930
1940
1950
1960
1970
1980
1990
2000
2010
+%
% ↓

Sample Means

```
> colMeans(merge(data.xts$MKT_RF,data.xts$SMB,data.xts$HML))
   MKT_RF       SMB       HML
0.6584201 0.2100091 0.3885571
```

→ sample mean

>0 >0

```
> # excess return for tech
> tech_rf = data.xts$HITEC-data.xts$RF

> colnames(tech_rf) <- c("tech_rf")
```

name the excess return

```
> # excess return for manuf
> manuf_rf = data.xts$MANUF-data.xts$RF

> colnames(manuf_rf) <- c("manuf_rf")
```

```
Call:
lm(formula = manuf_rf ~ data.xts$MKT_RF + data.xts$SMB + data.xts$HML)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.2149 -0.8313 -0.0042  0.7926  8.3613 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)     -0.01501    0.05484  -0.274    0.784    
data.xts$MKT_RF  1.03610    0.01095  94.599   <2e-16 ***
data.xts$SMB     0.66289    0.01791  37.022   <2e-16 ***
data.xts$HML     0.43358    0.01601  27.080   <2e-16 ***
```

$\hat{\beta}_2$ $\hat{\beta}_3$

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.794 on 1091 degrees of freedom
Multiple R-squared:  0.9395,	Adjusted R-squared:  0.9393 
F-statistic:  5646 on 3 and 1091 DF,  p-value: < 2.2e-16
```

Manufacturing portfolio

- The three factor model can be stated as

$$r_{Mt} - r_{ft} = \alpha_M + \beta_{M1}(r_{mt} - r_{ft}) + \beta_{M2}\text{SMB}_t + \beta_{M3}\text{HML}_t + \epsilon_{Mt}.$$

- The fitted equation for the manufacturing portfolios is given by

$$\widehat{r_{Mt} - r_{ft}} = -0.015 + 1.03(r_{mt} - r_{ft}) + \underbrace{0.66}_{>0}\text{SMB}_t + \underbrace{0.434}_{>0}\text{HML}_t.$$

- This portfolio tracks the market and $\boxed{\text{behaves}}$ more like a small and value portfolio.

- From the R^2 we see that these three factors explain 94% of variability of returns across this period.

```
Call:
lm(formula = tech_rf ~ data.xts$MKT_RF + data.xts$SMB + data.xts$HML)

Residuals:
    Min      1Q  Median      3Q     Max 
-15.686  -1.644  -0.233   1.235  35.656 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      0.20070    0.09359   2.144   0.0322 *  
data.xts$MKT_RF  1.21291    0.01869  64.888  < 2e-16 ***
data.xts$SMB     0.94741    0.03056  31.003  < 2e-16 ***
data.xts$HML    -0.18833    0.02733  -6.892 9.28e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.062 on 1091 degrees of freedom
Multiple R-squared:  0.8684,	Adjusted R-squared:  0.8681 
F-statistic:  2400 on 3 and 1091 DF,  p-value: < 2.2e-16
```

Technology portfolio

- Consider the technology portfolio.

- The fitted equation for this portfolio is given by

$$\widehat{r_{Tt} - r_{ft}} = 0.201 + 1.21(r_{mt} - r_{ft}) + 0.95\text{SMB}_t - 0.188\text{HML}_t.$$

>0 <0

- This portfolio is more aggressive than the market and $\boxed{\text{behaves}}$ more like a small and growth portfolio.

- From the $R^2 = 0.87$ we see that these three factors explain 87% of variability of returns across this period.

Multi-Factor CAPM

factor

- An additional factor called “Momentum” was also suggested by Carhart (1997).
- Momentum captures returns constructed by buying stocks with high returns and selling stocks with low returns over the same period.
- This factor captures hedging behavior of investors.

Multi-Factor CAPM

Using these additional factors we can construct the multi-factor CAPM:

$$r_{it} - r_{ft} = \alpha_i + \beta_{i1}(r_{mt} - r_{ft}) + \beta_{i2}\mathrm{SMB}_t + \beta_{i3}\mathrm{HML}_t + \beta_{i4}\mathrm{MOM}_t + \epsilon_{it}$$

4

H_0: If $\beta_{i2} = \beta_{i3} = \beta_{i4} = 0$ we revert to the simple CAPM.

We can test statistically the significance of these additional slope coefficients.

CAPM and Multi-Factor Regression Model Diagnostics

Diagnostics for the CAPM Model

- It is important that we have some diagnostic procedures to deduce whether the CAPM model is correctly specified.

- If CAPM is not correct, interpretations of β's and α are incorrect.

- Consider several types of model diagnostics:

  1. Model fit
  2. Diagnostics for the independent variables
  3. Diagnostics for the disturbance term

Model Fit

If a model fits the data well, it should be the case that the model explains a significant portion of variation in the data.

We use the coefficient of determination, or R^2 (where $0 \leq R^2 \leq 1$).

The adjusted R^2 penalizes the model for adding regressors which do not explain variability in the dependent variable.

Model Fit

For the manufacturing portfolio discussed earlier, the R^2 for the standard CAPM is $R^2 = 0.8166$ and the adjusted R^2, $\bar{R}^2$, for the multi-factor CAPM are $\bar{R}^2 = 0.9393$.

For the technology portfolio, the R^2 for the standard CAPM is $R^2 = 0.75$ and the adjusted R^2, $\bar{R}^2$, for the multi-factor CAPM are $\bar{R}^2 = 0.87$

These R^2 and $\bar{R}^2$ values indicate good model fit.

A high R^2 indicates that the multi-risk-factors (market portfolio, size, value, and momentum) are very important in explaining the variation in the stock portfolio.

CAPM - manufacturing

```
Call:
lm(formula = manuf_rf ~ data.xts$MKT_RF)

Residuals:
    Min      1Q  Median      3Q     Max 
-11.995  -1.584  -0.258   1.314  33.542 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      0.16455    0.09503   1.732   0.0836 .  
data.xts$MKT_RF  1.23068    0.01764  69.754   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.121 on 1093 degrees of freedom
Multiple R-squared:  0.8166,    Adjusted R-squared:  0.8164 
F-statistic:  4866 on 1 and 1093 DF,  p-value: < 2.2e-16
```

CAPM - technology

```
Call:
lm(formula = tech_rf ~ data.xts$MKT_RF)

Residuals:
    Min      1Q  Median      3Q     Max 
-16.193  -2.303  -0.278   1.664  42.303 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      0.22677    0.12862   1.763   0.0782 .  
data.xts$MKT_RF  1.36434    0.02388  57.138   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.224 on 1093 degrees of freedom
Multiple R-squared:  0.7492,	Adjusted R-squared:  0.749 
F-statistic:  3265 on 1 and 1093 DF,  p-value: < 2.2e-16
```

Borrowing from the CAPM, the pricing relation of the three-factor model is:

$$E\left(R\right) - r_f = \beta \times \left(\underbrace{E\left(R_M\right) - r_f}_{0.66}\right) + s \times \underbrace{E\left(SMB\right)}_{0.21} + h \times \underbrace{E\left(HML\right)}_{0.39}$$

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The risk premiums for the three factors can be estimated using the historical data.

Diagnostics for the Independent Variables

Diagnostics for the Independent Variable

- The CAPM model explains the behavior of risky assets in terms of their exposure to market risk.
- To establish the validity of this model we will rely on hypothesis testing.
- To determine the importance of the CAPM factors, we can test the parameters of the CAPM model.

Single Parameter Tests

- The importance of the market factor in the CAPM can be assessed by conducting a significance test of the market factor.
- Conduct a statistical test of the following **null and alternative hypotheses**

$$\mathrm{H}_0 : \beta_{i1} = 0 \text{ [Market factor is not priced]}$$
$$\mathrm{H}_1 : \beta_{i1} \neq 0 \text{ [Market factor is priced]}$$

Single Parameter Tests

- The **test statistic** to perform this test is the t-statistic, $t = \hat{\beta}_{i1}/\text{s.e.}(\hat{\beta}_{i1})$, which is asymptotically distributed as a normal random variable.

- In small samples it is distributed as Student-t with $T - K - 1$ degrees of freedom, where K is the number of regressors in the model.

- **The decision rule** is the standard decision rule: at the 5% level,

$$\begin{aligned} \text{if p-value} < .05 &: \text{ Reject } H_0 \\ \text{if p-value} > .05 &: \text{ Do not Reject } H_0 \end{aligned}$$

- Similarly, the t-test can be used to determine whether the α risk of the asset is significant.

Example: IBM

Consider that the CAPM model based on IBM, with the market portfolio represented via the CRSP (center for research in securities prices) data.

Using monthly data from 1980-2017, the predicted CAPM model is:

$$\widehat{y}_{IBM,t} = \hat{\alpha} + \hat{\beta}(r_{mt} - r_{ft})$$

$$= -0.175 + 0.8778(r_{mt} - r_{ft})$$

```
> head(data.xts)
         IBM_RET_RF MKT_RF
Jan 1980   8.1285714   5.51
Feb 1980  -8.3581239  -1.22
Mar 1980 -13.4147244 -12.90
Apr 1980  -2.8295067   3.97
May 1980   0.1011617   5.26
Jun 1980   5.4848081   3.06

> tail(data.xts)
         IBM_RET_RF MKT_RF
May 2017 -4.5898343   1.06
Jun 2017  0.4504705   0.78
Jul 2017 -5.6566662   1.87
Aug 2017 -1.4831062   0.16
Sep 2017  1.5675815   2.51
Oct 2017  5.9299516   2.25
```

% %

```
> fit <- lm(data.xts$IBM_RET_RF~data.xts$MKT_RF)

> summary(fit)

Call:
lm(formula = data.xts$IBM_RET_RF ~ data.xts$MKT_RF)

Residuals:
     Min       1Q   Median       3Q      Max
-25.7623  -3.6163  -0.4653   3.3772  29.9054

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     -0.17521    0.29349  -0.597    0.551
data.xts$MKT_RF  0.87782    0.06586  13.328   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.183 on 452 degrees of freedom
Multiple R-squared:  0.2821,	Adjusted R-squared:  0.2805
F-statistic: 177.6 on 1 and 452 DF,  p-value: < 2.2e-16
```

α %

→ slide 62

%

Example

Let us now conduct a statistical test to determine whether or not IBM tracks the market. This would require us to test, in the single-factor CAPM model,

$$\begin{aligned} H_0 &: \beta_{IBM,1} = 1 \Rightarrow \text{fail to reject} \\ H_1 &: \beta_{IBM,1} \neq 1 \end{aligned}$$

The test statistic is

$$t = \frac{\hat{\beta}_{IBM,1} - 1}{\text{s.e.}(\hat{\beta}_{IBM,1})} = \frac{0.87782 - 1}{.0659} = -1.854$$

This yields a p-value of $p > .05$. Therefore, we cannot reject the null hypothesis at the 5% significance level.

Example

We can also conduct a statistical test of whether IBM yields an **adequate reward for assumed risk**. This would require us to test:

$$
\begin{aligned}
H_0 &: \alpha_{IBM} = 0 \\
H_1 &: \alpha_{IBM} \neq 0
\end{aligned}
$$

fail to reject.

The test statistic is

$$t = \frac{\hat{\alpha}_{IBM} - 0}{\text{s.e.}(\hat{\alpha}_{IBM})} = \frac{-0.175}{0.293} = -0.597.$$

This yields a p-value with $p > .05$. Conclude that IBM yields an **adequate reward for assumed risk**.

Joint Parameter Tests

- Often we wish to understand if the other factors in the CAPM are needed.

- This requires a joint hypothesis test on all the additional factors.

- It leads to the following hypotheses:

$$
\begin{aligned}
H_0 &: \beta_{i2} \overset{1}{=} \beta_{i3} \overset{2}{=} \beta_{i4} \overset{3}{=} 0 \Rightarrow \text{factors are useless} \\
H_1 &: \text{ at least one is non-zero}
\end{aligned}
$$

- In general, there can be l general restrictions being tested, but in this case we are only testing $l = 3$ restrictions.

d.f. $\Rightarrow \chi^2$

Joint Parameter Tests

- A test of H_0 requires estimating both models and comparing the results.

- Recall : **residual sum of squares**, RSS,

$$\mathrm{RSS}_i = \sum_{t=1}^{T} \hat{\epsilon}_{it}^2.$$

- Denote RSS_{i0} for RSS under standard CAPM and RSS_{i1} for RSS of multi-factor CAPM model applied to excess returns of asset i.

- The **test statistic** is

CAPM (0) muti factor (1)

$$\mathrm{J} = \frac{\mathrm{RSS}_{i0} - \mathrm{RSS}_{i1}}{(\mathrm{RSS}_{i1}/(T - \underbrace{K}_{\text{\# of regressor for muti factor.}} - 1))}.$$

(1)

- J is asymptotically χ_l^2 with degrees of freedom parameter equal to the number of restrictions being tested ($l = 3$ in this case).

CAPM - technology portfolio

```
Call:
lm(formula = tech_rf ~ data.xts$MKT_RF)

Residuals:
    Min      1Q  Median      3Q     Max 
-16.193  -2.303  -0.278   1.664  42.303 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      0.22677    0.12862   1.763   0.0782 .  
data.xts$MKT_RF  1.36434    0.02388  57.138   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                                d.f
Residual standard error: 4.224 on 1093 degrees of freedom
Multiple R-squared:  0.7492,	Adjusted R-squared:  0.749 
F-statistic:  3265 on 1 and 1093 DF,  p-value: < 2.2e-16
```

3-factor - technology portfolio

```
> fit.tech <- lm(tech_rf~data.xts$MKT_RF+data.xts$SMB+data.xts$HML)

> summary(fit.tech)

Call:
lm(formula = tech_rf ~ data.xts$MKT_RF + data.xts$SMB + data.xts$HML)

Residuals:
    Min      1Q  Median      3Q     Max 
-15.686  -1.644  -0.233   1.235  35.656 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      0.20070    0.09359   2.144   0.0322 *  
data.xts$MKT_RF  1.21291    0.01869  64.888  < 2e-16 ***
data.xts$SMB     0.94741    0.03056  31.003  < 2e-16 ***
data.xts$HML    -0.18833    0.02733  -6.892 9.28e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
                                                  T-K-1 => d.f.
Residual standard error: 3.062 on 1091 degrees of freedom
Multiple R-squared:  0.8684,	Adjusted R-squared:  0.8681 
F-statistic:  2400 on 3 and 1091 DF,  p-value: < 2.2e-16
```

Let's return to the 3-factor CAPM estimated on the technology portfolio. We'll test if that model reduces to the standard CAPM. **The hypotheses**:

$$H_0 : \beta_{i2} = \beta_{i3} = 0$$
$$H_1 : \text{ at least one is non-zero}$$

$RSS = \sum_{i=1}^{T} \hat{e}_i^2$

Recall

$$SER = \sqrt{\frac{1}{T-K-1} RSS}$$

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Under CAPM

$$4.224 = \sqrt{\frac{RSS_0}{1093}}$$
$$RSS_0 = 19501$$

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d.f

and under FF3F

$$3.062 = \sqrt{\frac{RSS_1}{1091}}$$
$$RSS_1 = 10229$$

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In this case, we have $RSS_{i0} = 19501$, $RSS_{i1} = 10229$ and **the test statistic** is

if about the same factors are useless.

$$J = \frac{RSS_0 - RSS_1}{RSS_1/(T-K-1)}$$

$$= \frac{19501 - 10229}{10229/1091} = 988.93$$

We can then compare J to a χ^2_2 critical value. The value of J is much larger than that of the χ^2_2 critical value at any conventional level of significance.

reject Ho.

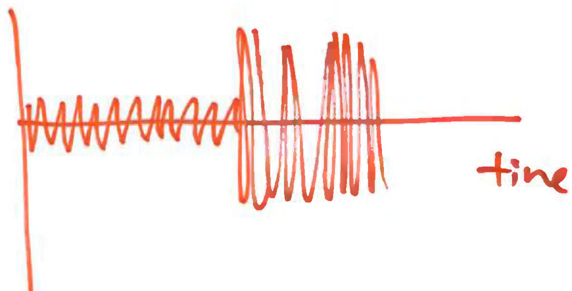
Diagnostics for the Disturbance Term

$E(\varepsilon_t) = 0$

$$t = \frac{\hat{\beta}_j}{s.e.(\hat{\beta}_j)}$$

$s.e.(\hat{\beta}_j) \downarrow \Rightarrow \hat{\beta}_j$ more precise estimation of true β.

Assumption ① $E(\varepsilon_i^2) = \sigma^2$ a number.



② $E(\underset{\text{today}}{\underset{X}{\varepsilon_t}}\ \underset{\text{yesterday}}{\underset{Y}{\varepsilon_{t-1}}}) = 0$ NO autocorrelation $\rightarrow$ forget

$cov(X, Y)$